AlphaVerify: Mathematical Specification

Measurement version `shared-outcome-cache-float64-v2`

Contents

1 Purpose and scope

AlphaVerify measures how conditioning on a feature changes the probability of touching signed return barriers over forward horizons. Its basic computational unit is a complete barrier–horizon Surface. The four stages measure probability surfaces, subtract a market baseline, validate a score of each resulting shift surface against synthetic histories, and retain the observed surfaces that clear the test.

This document defines those transformations and their numerical contract. It is intended for readers implementing the method; no particular array library is assumed. Primitive operations and notation are defined below. The same measurement and scoring rules apply to observed and synthetic histories. Section 8 collects array layouts, artifact references, and cache rules; these determine how results are represented and reused without changing the mathematical operations.

What the comparison establishes

The current decision is an exploratory comparison with a fitted independent Gaussian OHLC simulator. A small Monte Carlo i-value means that this simulator rarely produces a score as large as the observed one. It does not by itself establish incremental predictive information, out-of-sample performance, or a calibrated 5% false-positive rate in markets with temporal dependence. Model limitations and support conditioning are part of the interpretation of the decision, not merely implementation details (Section 6). This specification describes the current algorithm; explicitly identified diagnostics and alternatives are proposals, not claims about implemented behavior.

2 Central concepts

2.1 The Surface as the computational unit

Fix an ordered signed-barrier grid Δ and an ordered forward-horizon grid Θ. Their elements are the two coordinates of a Surface:

- $\delta \in \Delta$ is a signed return barrier, expressed as a fraction of the starting date's close. For example, $\delta = -0.05$ is a barrier 5% below that close, and $\delta = +0.05$ is a barrier 5% above it.
- $\theta \in \Theta$ is a forward horizon measured in subsequent observations (bars). For example, $\theta = 10$ asks whether the barrier is touched within the next ten bars, rather than only at the end of the tenth bar.

Thus Δ and Θ are the grids, while δ and θ select one barrier and one horizon within them. A real-valued Surface is a function on their product:

The Surface space

$$\mathcal{S} = \mathbb{R}^{\Delta\times\Theta}, \qquad F \in \mathcal{S}, \qquad F(\delta, \theta) \in \mathbb{R}.$$

Thus $\mathcal{S}$ is the space of surfaces, and F is one Surface. Its array representation has $|\Delta|$ rows and $|\Theta|$ columns. The entire grid is one element—one point in this space—even though its construction and evaluation use individual coordinates.

Partial support

A complete Surface retains the full grid even when some entries are unsupported. Formally, a partially supported Surface is represented by an element $F \in \mathcal{S}$ together with a Boolean support mask on $\Delta \times \Theta$. Values outside the support are arbitrary and never used. Numerical arrays may store non-finite values there instead; NaN is a storage convention, not an element of $\mathbb{R}$. Completeness describes the grid, not a guarantee that every entry is available.

2.2 Paths

The path index $i = 0, 1, \ldots, I$ identifies one complete OHLCV price history. Path $i = 0$ is the observed market history. Paths $i = 1, \ldots, I$ are random synthetic histories generated from the fitted null model for hypothesis testing; the current configuration uses $I = 1{,}000$. Each path is measured and scored by the same procedure, so its surfaces and scores can be compared with those of the observed history. Section 6.3 defines how these null histories are generated.

We write the path label as a parenthesized superscript (i), and the within-path date or bin as a subscript. The parentheses distinguish the path label from an exponent. For example, $X_t^{(i)}$ is the outcome Surface for path i and date t, whereas $Y_k^{(i)}$ is the conditional probability Surface for path i and bin k. A date-level outcome and a bin-level probability do not have the same outer indices: aggregation replaces a collection indexed by dates with one indexed by bins.

2.3 The objects carried through the pipeline

All four Surface objects belong to the same ambient space:

Four Surface objects, one common space

$$X_t^{(i)}, \quad Y_k^{(i)}, \quad B^{(i)}, \quad Z_k^{(i)} \in \mathcal{S},$$

with the support-mask convention above wherever needed. Their outer indices identify different members of $\mathcal{S}$; their common internal coordinates are always (δ, θ). On supported coordinates:

Outcome Surface

$X_t^{(i)}(\delta, \theta) \in \{0, 1\}$ records whether the signed barrier δ was touched within horizon θ after date t: 0 means not touched and 1 means touched.

Conditional probability Surface

$Y_k^{(i)}(\delta, \theta) \in [0, 1]$ gives the measured probability of that touch among eligible dates in condition bin k. The interval $[0, 1]$ corresponds to 0%–100%; for example, 0.60 means a 60% touch probability.

Unconditional probability Surface (baseline Surface)

$B^{(i)}(\delta, \theta) \in [0, 1]$ gives the unconditional probability of the same touch, using every price-eligible date in path i, regardless of feature values or bin membership. It uses the same probability scale: 0.40 means 40%. There is one unconditional probability Surface per path, shared by all condition bins.

Shift Surface

$Z_k^{(i)}(\delta, \theta) = Y_k^{(i)}(\delta, \theta) - B^{(i)}(\delta, \theta)$ is the change in touch probability, as a probability difference. Its range is $[-1, 1]$: positive means more likely to touch, negative means less likely, and zero means the same probability as the baseline. For example, 0.60 conditional versus 0.40 baseline gives +0.20, displayed as +20%: a 20-percentage-point difference, not a relative percentage change.

These different value ranges and meanings do not change the ambient space.

The baseline is the unrestricted case of the same averaging operation that produces conditional probability surfaces. Writing $\Pi^{(i)}[A]$ for the probability Surface measured over a date group A, we have

$$Y_k^{(i)} = \Pi^{(i)}[A_k^{(i)}], \qquad B^{(i)} = \Pi^{(i)}[\mathcal{T}],$$

where $A_k^{(i)}$ contains the dates assigned to bin k, and $\mathcal{T}$ contains all observation dates. Horizon-specific price eligibility applies in both cases. Section 4.3 defines this operator precisely. The baseline is not bin 0: that position is already the lowest feature bin and requires a finite feature value.

Object	**One Surface per**	**Meaning of an entry** (δ, θ)
$X_t^{(i)}$	Path and date	Whether the barrier was touched
$Y_k^{(i)}$	Path and bin	Conditional touch probability
$B^{(i)}$	Path	Unconditional touch probability
$Z_k^{(i)}$	Path and bin	Conditional minus baseline

2.4 The scalar score and its validity flag

Scoring assigns two scalar quantities to each path i and condition bin k:

One scalar score and one validity flag per bin

$$Q_k^{(i)} \in \mathbb{R}_{\geq 0} \subset \mathbb{R}, \qquad V_k^{(i)} \in \{0, 1\}.$$

Their outer indices identify a path and bin, just as for $Z_k^{(i)}$. They have no internal barrier or horizon coordinates: each summarizes one complete shift Surface.

Score

$Q_k^{(i)}$ is a nonnegative real number obtained by summing weighted contributions from usable coordinates of $Z_k^{(i)}$. It measures the magnitude of the difference between shifts at opposite signed barriers. It is not a probability and can exceed one.

Validity flag

$V_k^{(i)}$ is a Boolean indicator encoded as 0 or 1. It equals 1 when the score has at least one usable coordinate, and 0 when it has none. An unsupported score is stored as $Q_k^{(i)} = 0$ with $V_k^{(i)} = 0$; a supported score may also be zero, with $V_k^{(i)} = 1$.

Validity records support, not statistical significance. Section 6.1 defines the contributions and support rule; Stage 3 compares supported observed scores with scores from synthetic histories.

The central computation

$$\{X_t^{(i)}\}_{t\in\mathcal{T}} \xrightarrow{\text{eligible averaging}} (\{Y_k^{(i)}\}_k, B^{(i)}) \xrightarrow{\text{baseline subtraction}} \{Z_k^{(i)}\}_k \xrightarrow{\text{scoring}} \{(Q_k^{(i)}, V_k^{(i)})\}_k.$$

Averaging uses horizon-dependent eligible dates, so different columns need not have the same denominator.

Stage 1 constructs the observed probability surfaces. Stage 2 constructs their shifts $Z_k^{(0)}$. Stage 3 scores the observed shifts and repeats the computation on synthetic histories to obtain a null comparison for each bin. Stage 4 selects and renders the complete observed shift surfaces whose bins clear that comparison.

The Surface is the unit being scored and selected. Its score also depends on support in the other bins of the same feature: a barrier pair contributes only when at least two bins are usable at that horizon. This dependency is made explicit in Section 6.1.

3 Mathematical objects and conventions

3.1 Outer indices and Surface coordinates

The exposition fixes one feature (one condition node). The computation is repeated for every node; feature-dependent quantities below implicitly belong to that fixed node. Outcome surfaces and market baselines are shared across nodes within a path.

The path label always appears as a parenthesized superscript; subscripts identify dates, bins, or grid coordinates within that path. When a quantity also has a descriptive superscript, both labels are retained, for example $N_{k,\theta}^{\text{condition},(i)}$ for conditional observation counts and $R_{t,\theta}^{-,(i)}$ for downside excursions.

Symbol	Meaning
$i \in \{0, \dots, I\}$	Path label; 0 is observed and $I = 1{,}000$ paths are synthetic.
$t \in \mathcal{T}$	identifies an observation date. In forward expressions, $t + j$ means the jth subsequent observation.
$x_t^{(i)}$	is the feature value used to assign date t to a bin.
$k \in \{0, \dots, K_{\text{req}} - 1\}$	is a requested bin position. Path i has $K^{(i)} \le K_{\text{req}}$ effective bins, occupying positions $0, \dots, K^{(i)} - 1$.
$\delta \in \Delta,\ \theta \in \Theta$	Signed return barrier and forward horizon in bars; grids are ordered.
$\Delta_\pm$	Nonzero grid rows with unique opposite-sign partners.
$e^{(i)},\ \kappa_t^{(i)},\ A_k^{(i)}$	Ordered bin edges, date assignment, and dates in bin k.
$\mathcal{E}_\theta^{(i)},\ \mathcal{E}_{k,\theta}^{(i)}$	Price-eligible dates, and their intersection with bin k.
$N^{\text{condition}},\ N^{\text{baseline}},\ N^{\text{hit}}$	Conditional sample count, baseline sample count, and touch count; $n_{\min} = 30$.

Symbol	Meaning
$X_t^{(i)}, Y_k^{(i)}, B^{(i)}, Z_k^{(i)}$	Outcome, conditional probability, baseline probability, and shift Surfaces.
$\Pi^{(i)}[A]$	Eligible averaging of outcome Surfaces over date group A.
$R_{t,\theta}^{-,(i)}, R_{t,\theta}^{+,(i)}$	Minimum downside and maximum upside forward excursions.
$\mathcal{L}^{(i)},\ \mathcal{U}_k^{(i)}$	Locally eligible bins and usable scoring coordinates.
$w_\delta,\ H_k^{(i)},\ Q_k^{(i)},\ V_k^{(i)}$	Barrier weight, contribution Surface, scalar score, and support flag.
$\widehat{i}_k,\ C_k, \mathcal{K}_{\text{sel}}$	Monte Carlo i-value, clearing flag, and selected bin positions.
$g_t^{(i)}, z_t^{(i)}, \mu, \Sigma$	OHLC log components, standard Gaussian innovations, fitted mean, and covariance.
$\epsilon, L_\Sigma, I_4, C_{\text{init}}$	Covariance jitter, lower Cholesky factor, four-dimensional identity, and initial close.
j, γ	Local dummy indices: j indexes offsets, edges, components, or time within its formula; γ indexes barriers.

Bin positions are zero-based, matching the insertion-index convention used in the implementation. Bin k refers to an ordered position within each path's binning; its numerical feature boundaries can differ between paths.

Primitive operations

$\mathbf{1}[P]$ is 1 when proposition P is true and 0 otherwise. For a positive-definite matrix A, $\text{chol}(A)$ is the lower triangular matrix L with positive diagonal satisfying $LL^\mathsf{T} = A$. For an increasing edge vector e, $\text{searchsorted}(e, x, \text{left}) = |\{r : e_r < x\}|$; equality therefore enters the lower bin. For example, edges $(2, 5)$ assign $2, 3, 5, 6$ to bins $0, 1, 1, 2$, respectively.

Quantiles use linear interpolation of the sorted finite sample $a_0 \leq \cdots \leq a_{n-1}$. For probability $u \in [0, 1]$, let $h = (n-1)u$, $r = \lfloor h \rfloor$, and $s = \lceil h \rceil$; then $\text{quantile}_u(a) = a_r + (h - r)(a_s - a_r)$. An empty finite sample has no finite edges and assigns no dates to any bin. NaN denotes an unavailable stored value, never a numeric zero.

3.2 Values, counts, and support

Outcome entries are Boolean on price-eligible coordinates; probability entries lie in $[0, 1]$ where supported; shift entries are probability differences in $[-1, 1]$. The support threshold is

$$n_{\min} = 30.$$

Counts depend on horizon because fewer dates have longer forward windows. Probability and shift surfaces may therefore be only partially supported.

A support mask is separate from a value. In particular, an unsupported score is stored as zero with a false flag; a supported score can also equal zero. All sums below use eligible or usable index sets so that excluded non-finite values are never interpreted as products such as $0 \cdot \text{NaN}$.

4 Stage 1: measurement of probability surfaces

Measurement turns each history into date-level outcome surfaces, then averages those outcomes over all eligible dates and over feature bins. Its outputs are conditional probability surfaces $Y_k^{(i)}$, an unconditional probability Surface $B^{(i)}$, and the counts supporting them. Stage 1 retains the observed outputs $(i = 0)$; Stage 3 applies the same definitions to null paths.

4.1 From prices to an outcome Surface

For every path, starting date, and horizon, define the greatest downside and upside excursion reached by future intrabar prices:

$$R_{t,\theta}^{-,(i)} = \frac{\min_{1\le j\le\theta} \mathrm{Low}_{t+j}^{(i)}}{\mathrm{Close}_t^{(i)}} - 1, \tag{1}$$

$$R_{t,\theta}^{+,(i)} = \frac{\max_{1\le j\le\theta} \mathrm{High}_{t+j}^{(i)}}{\mathrm{Close}_t^{(i)}} - 1. \tag{2}$$

Dates without θ future observations have non-finite excursions. Define the price-eligible date set

$$\mathcal{E}_\theta^{(i)} = \{t \in \mathcal{T} : R_{t,\theta}^{-,(i)} \text{ and } R_{t,\theta}^{+,(i)} \text{ are finite}\}.$$

For eligible dates, the outcome Surface has entries

$$X_t^{(i)}(\delta,\theta) = \begin{cases} \mathbf{1}[R_{t,\theta}^{-,(i)} \le \delta], & \delta < 0, \\ \mathbf{1}[R_{t,\theta}^{+,(i)} \ge \delta], & \delta \ge 0. \end{cases}$$

An entry answers: starting from this date's close, was the signed barrier reached by a future low or high within this horizon? The zero barrier uses the upside branch. It belongs to the measured grid but is excluded from the paired-barrier score.

$X_t^{(i)}$ depends only on the history and grids. Feature bins select which outcome surfaces to average; they do not change the outcome surfaces.

4.2 From feature values to date groups

For a continuous feature, request the interior quantile edges

$$e_j^{(i)} = \mathrm{quantile}_{j/K_{\mathrm{req}}}(\{x_t^{(i)} : x_t^{(i)} \text{ is finite}\}), \qquad j = 1,\dots,K_{\mathrm{req}} - 1.$$

Remove non-finite and duplicate edges, and retain only edges satisfying $\min(x^{(i)}) < e_j^{(i)} \le \max(x^{(i)})$, with extrema taken over finite feature values. Write $e^{(i)}$ for the resulting ordered edge vector and $K^{(i)} = |e^{(i)}| + 1$. Heavy ties can reduce the effective bin count; an edge at the maximum can leave the final bin empty.

For finite feature values, assign

$$\kappa_t^{(i)} = \mathrm{searchsorted}(e^{(i)}, x_t^{(i)}, \mathrm{left}).$$

An observation equal to an edge enters the lower bin. Define the date group for bin k and its price-eligible subset at horizon θ by

$$A_k^{(i)} = \{t \in \mathcal{T} : x_t^{(i)} \text{ is finite, } \kappa_t^{(i)} = k\}, \qquad \mathcal{E}_{k,\theta}^{(i)} = A_k^{(i)} \cap \mathcal{E}_\theta^{(i)}.$$

Positions $k \ge K^{(i)}$ have no eligible dates.

4.3 Averaging outcome surfaces

For any date group $A \subseteq \mathcal{T}$, define the probability Surface

$$\Pi^{(i)}[A](\delta,\theta) = \frac{\sum_{t\in A\cap\mathcal{E}_\theta^{(i)}} X_t^{(i)}(\delta,\theta)}{|A\cap\mathcal{E}_\theta^{(i)}|}, \qquad |A\cap\mathcal{E}_\theta^{(i)}| \ge n_{\min}.$$

Entries are non-finite when the count requirement fails. Conditional measurement and baseline measurement are two applications of this operator:

$$Y_k^{(i)} = \Pi^{(i)}[A_k^{(i)}], \qquad B^{(i)} = \Pi^{(i)}[\mathcal{T}].$$

Thus the baseline is the special case with no restriction on the date group; it still enforces price eligibility and the minimum count.

To make the supporting counts explicit, define

$$N_{k,\theta}^{\text{condition},(i)} = |\mathcal{E}_{k,\theta}^{(i)}|, \tag{3}$$

$$N_k^{\text{hit},(i)}(\delta,\theta) = \sum_{t\in\mathcal{E}_{k,\theta}^{(i)}} X_t^{(i)}(\delta,\theta), \tag{4}$$

$$N_\theta^{\text{baseline},(i)} = |\mathcal{E}_\theta^{(i)}|. \tag{5}$$

The probability surfaces are

$$Y_k^{(i)}(\delta,\theta) = \frac{N_k^{\text{hit},(i)}(\delta,\theta)}{N_{k,\theta}^{\text{condition},(i)}}, \qquad N_{k,\theta}^{\text{condition},(i)} \geq n_{\min}, \tag{6}$$

$$B^{(i)}(\delta,\theta) = \frac{\sum_{t\in\mathcal{E}_\theta^{(i)}} X_t^{(i)}(\delta,\theta)}{N_\theta^{\text{baseline},(i)}}, \qquad N_\theta^{\text{baseline},(i)} \geq n_{\min}. \tag{7}$$

When the stated count requirement fails, the corresponding entries are non-finite.

The baseline includes every price-eligible date, independently of feature warm-up or missing feature values. The conditional surface uses only eligible dates in its bin. Thus the baseline generally cannot be recovered by averaging the conditional surfaces over bins. Each synthetic path has its own baseline $B^{(i)}$.

4.4 Worked example: measuring a Surface

Take five barriers and seven daily horizons:

$$\Delta = \{-2\%, -1\%, 0\%, +1\%, +2\%\}, \qquad \Theta = \{1, 2, 3, 4, 5, 6, 7\}.$$

The percentage labels stand for returns $-0.02, -0.01, 0, 0.01, 0.02$. Assume a market with a bar every calendar day, so seven bars cover one week. For a trading-day-only history, these horizons instead cover seven trading days.

Start from a close of 100. The five barrier prices are then $98, 99, 100, 101, 102$. Suppose future intrabar prices first reach 100 on day 1, 101 on day 2, 99 on day 3, 102 on day 5, and 98 on day 6. This gives the following outcome Surface:

$$X_t^{(i)} = \begin{array}{c|ccccccc} \delta\backslash\theta & 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ \hline -2\% & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ -1\% & 0 & 0 & 1 & 1 & 1 & 1 & 1 \\ 0\% & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ +1\% & 0 & 1 & 1 & 1 & 1 & 1 & 1 \\ +2\% & 0 & 0 & 0 & 0 & 1 & 1 & 1 \end{array}.$$

For example, $X_t^{(i)}(+2\%, 4) = 0$ but $X_t^{(i)}(+2\%, 5) = 1$: the price had not reached 102 within four days, but had within five. Once a row becomes 1, it stays 1 at longer horizons, even if price later reverses. The zero row records a future high at or above the starting close; it is not automatically a touch at the starting date.

Now collect 100 starting dates assigned to the same condition bin k, each with seven eligible future days. Each date contributes one such 5×7 outcome Surface. Suppose their coordinatewise averages are

$$Y_k^{(i)} = \begin{array}{c|ccccccc} \delta\backslash\theta & 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ \hline -2\% & 0.05 & 0.10 & 0.15 & 0.20 & 0.25 & 0.30 & 0.35 \\ -1\% & 0.15 & 0.20 & 0.25 & 0.30 & 0.35 & 0.40 & 0.45 \\ 0\% & 0.70 & 0.75 & 0.80 & 0.85 & 0.90 & 0.95 & 1.00 \\ +1\% & 0.25 & 0.30 & 0.35 & 0.40 & 0.45 & 0.50 & 0.55 \\ +2\% & 0.15 & 0.20 & 0.25 & 0.30 & 0.35 & 0.40 & 0.45 \end{array}.$$

The entry $Y_k^{(i)}(+2\%, 5) = 0.35$ means that 35 of the 100 dates were followed by a touch of the +2% barrier within five days. The hit count is 35, the observation count is 100, and the measured probability is $35/100 = 35\%$. All 35 entries together form one conditional probability Surface. Real histories can have different observation counts at different horizons.

5 Stage 2: comparison with the baseline

Comparison converts a conditional probability Surface into a baseline-relative shift Surface.

5.1 Baseline subtraction

Define the shift Surface by

$$Z_k^{(i)} = (Y_k^{(i)} - B^{(i)}).$$

Subtraction is coordinatewise on their common support:

$$Z_k^{(i)}(\delta, \theta) = (Y_k^{(i)}(\delta, \theta) - B^{(i)}(\delta, \theta)).$$

A positive entry means that the barrier was touched more often in the condition bin than in the path's unconditional market history. A negative entry means it was touched less often. Null histories use their own $B^{(i)}$, never the observed baseline $B^{(0)}$.

5.2 Worked example: subtracting the baseline

Continue with the same five barriers, seven daily horizons, and conditional Surface $Y_k^{(i)}$ from Section 4.4. Suppose averaging over all price-eligible dates in the same path gives the baseline

$$B^{(i)} = \begin{array}{c|ccccccc} \delta\backslash\theta & 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ \hline -2\% & 0.15 & 0.20 & 0.25 & 0.30 & 0.35 & 0.40 & 0.45 \\ -1\% & 0.25 & 0.30 & 0.35 & 0.40 & 0.45 & 0.50 & 0.55 \\ 0\% & 0.70 & 0.75 & 0.80 & 0.85 & 0.90 & 0.95 & 1.00 \\ +1\% & 0.15 & 0.20 & 0.25 & 0.30 & 0.35 & 0.40 & 0.45 \\ +2\% & 0.05 & 0.10 & 0.15 & 0.20 & 0.25 & 0.30 & 0.35 \end{array}.$$

Subtract each baseline entry from the conditional entry at the same barrier and horizon. The result is one complete shift Surface, with every entry below on the probability-difference scale:

$$Z_k^{(i)} = (Y_k^{(i)} - B^{(i)}) = \begin{array}{c|rrrrrrr} \delta\backslash\theta & 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ \hline -2\% & -0.10 & -0.10 & -0.10 & -0.10 & -0.10 & -0.10 & -0.10 \\ -1\% & -0.10 & -0.10 & -0.10 & -0.10 & -0.10 & -0.10 & -0.10 \\ 0\% & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ +1\% & +0.10 & +0.10 & +0.10 & +0.10 & +0.10 & +0.10 & +0.10 \\ +2\% & +0.10 & +0.10 & +0.10 & +0.10 & +0.10 & +0.10 & +0.10 \end{array}.$$

For example, at $(+2\%, 5)$ the conditional probability is 35% and the baseline is 25%:

$$Z_k^{(i)}(+2\%, 5) = 0.35 - 0.25 = +0.10.$$

At $(-2\%, 5)$, the corresponding subtraction is $0.25 - 0.35 = -0.10$. The zero-barrier probabilities are equal, so their shifts are zero. The condition is associated with fewer downside touches and more upside touches across the seven horizons. Stage 2 retains the observed $Z_k^{(0)}$ together with references to its measurement artifacts.

6 Stage 3: validation by null comparison

Validation asks how often the specified synthetic-history model produces a bin score at least as large as the observed score. It first reduces each shift Surface to a scalar, then repeats measurement and scoring on null histories, and finally compares scores at the same bin position.

6.1 Scoring a shift Surface

Scoring keeps the full signed-barrier–by–horizon grid. It compares each shift with the shift at the opposite barrier, assigns a nonnegative contribution to each original coordinate, and sums that contribution Surface to obtain one bin score.

Reflection and weights

Let $\Delta_\pm$ contain the nonzero barriers δ for which the grid has exactly one δ row and exactly one $-\delta$ row. Define reflection on the original grid by

$$(\mathrm{Ref}\, F)(\delta,\theta) = \begin{cases} F(-\delta,\theta), & \delta \in \Delta_\pm, \\ F(\delta,\theta), & \delta \notin \Delta_\pm. \end{cases}$$

Reflection reorders rows without changing the Surface shape. When $\Delta_\pm \neq \varnothing$, assign the magnitude weights

$$w_\delta = \begin{cases} \dfrac{|\delta|}{\max_{\gamma\in\Delta_\pm} |\gamma|}, & \delta \in \Delta_\pm, \\ 0, & \delta \notin \Delta_\pm. \end{cases}$$

If $\Delta_\pm$ is empty, all weights are zero.

The current scorer silently ignores unmatched or ambiguously matched rows; it returns zero scores and false support flags when no pairs remain. This is a configuration failure mode, not evidence of no signal. Recommended diagnostics are paired and excluded row counts and an error when no pairs remain; these are not currently enforced. Numerically, the implementation matches using `isclose` with absolute tolerance 10^{-12} and relative tolerance 10^{-5}, and ignores magnitudes at most 10^{-12}. The exact reflection above assumes an unambiguous symmetric grid.

Usable coordinates

Support is determined on this same grid. First identify locally eligible bins at each nonzero matched barrier and horizon:

$$\mathcal{L}^{(i)}(\delta,\theta) = \left\{ k \in \{0,\ldots,K_{\text{req}}-1\} : \begin{array}{l} Z_k^{(i)}(\delta,\theta), Z_k^{(i)}(-\delta,\theta) \text{ are finite,} \\ N_{k,\theta}^{\text{condition},(i)} \geq n_{\min} \end{array} \right\}, \qquad \delta \in \Delta_\pm.$$

The usable coordinates for bin k are

$$\mathcal{U}_k^{(i)} = \{(\delta,\theta) \in \Delta_\pm \times \Theta : k \in \mathcal{L}^{(i)}(\delta,\theta),\ |\mathcal{L}^{(i)}(\delta,\theta)| \geq 2\}.$$

Thus support still depends on the other bins of the same feature.

Contribution and score

Define the full contribution Surface $H_k^{(i)} \in \mathcal{S}$ by

$$H_k^{(i)}(\delta,\theta) = \begin{cases} \dfrac{w_\delta}{2} \left| Z_k^{(i)}(\delta,\theta) - (\mathrm{Ref}\, Z_k^{(i)})(\delta,\theta) \right|, & (\delta,\theta) \in \mathcal{U}_k^{(i)}, \\ 0, & \text{otherwise.} \end{cases}$$

Both signed rows carry their own contribution. The factor $1/2$ accounts for the same absolute difference appearing at both δ and $-\delta$, preserving the score's scale. No reduced magnitude–horizon Surface is constructed. Unsupported coordinates, zero barriers, and unmatched barriers contribute zero.

The full-grid score and its validity flag are

$$Q_k^{(i)} = \sum_{\theta\in\Theta} \sum_{\delta\in\Delta} H_k^{(i)}(\delta,\theta), \qquad V_k^{(i)} = \mathbf{1}[\mathcal{U}_k^{(i)} \neq \varnothing].$$

Equal shifts at opposite barriers contribute zero; the absolute value discards the direction of the asymmetry. Larger barrier magnitudes receive greater linear weight. The sum is not normalized by the number of usable coordinates. It is a weighted probability-difference score, not a probability, and can exceed one. An unsupported bin has score zero and false validity; a supported zero score remains valid.

There is no inverse-variance weighting or normalization for usable grid size. Coordinates supported by 31 and 3,000 dates receive the same barrier weight. Under independent Bernoulli observations, the standard error at probability 0.5 with 30 dates is $\sqrt{0.25/30} \approx 0.091$. This is not an uncertainty bound here: forward windows overlap, the baseline overlaps each bin, and opposite-barrier outcomes are dependent. Precision weighting would need these covariances. Equal scoring rules do not compensate for a mismatch in observed and simulated dependence or support.

6.2 Worked example: scoring the shift

For the running example, take the Stage 2 Surface as the observed Surface $Z_k^{(0)}$. Its two negative-barrier rows are -0.10 and its two positive-barrier rows are $+0.10$ at every horizon. The bin has 100 eligible dates at every horizon; assume at least one other bin is locally eligible at both magnitudes and all seven horizons. Each nonzero coordinate is therefore usable.

The 1% rows have weight $0.01/0.02 = 0.5$, so each entry contributes $\frac{1}{2}(0.5)|-0.10-(+0.10)| = 0.05$. The 2% rows have weight 1, so each entry contributes $\frac{1}{2}(1)|-0.10-(+0.10)| = 0.10$. The resulting contribution Surface has the same 5×7 layout as the shift Surface:

$$H_k^{(0)} = \begin{array}{c|ccccccc} \delta\backslash\theta & 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ \hline -2\% & 0.10 & 0.10 & 0.10 & 0.10 & 0.10 & 0.10 & 0.10 \\ -1\% & 0.05 & 0.05 & 0.05 & 0.05 & 0.05 & 0.05 & 0.05 \\ 0\% & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ +1\% & 0.05 & 0.05 & 0.05 & 0.05 & 0.05 & 0.05 & 0.05 \\ +2\% & 0.10 & 0.10 & 0.10 & 0.10 & 0.10 & 0.10 & 0.10 \end{array}.$$

Each column contributes $0.10 + 0.05 + 0 + 0.05 + 0.10 = 0.30$. Summing the entire Surface gives

$$Q_k^{(0)} = \underbrace{0.30 + 0.30 + 0.30 + 0.30 + 0.30 + 0.30 + 0.30}_{\text{seven horizons}} = 2.10, \qquad V_k^{(0)} = 1.$$

Stage 3 compares this single score with scores from null histories; it does not test each of the 35 Surface entries separately.

Worked boundary cases: support is not a score

Use one horizon and $\Delta = (-0.02, +0.02)$, so both weights are 1. With two eligible bins, shifts $(-0.10, +0.10)$ in one bin give two contributions of 0.10, hence $Q = 0.20, V = 1$. Vary this setup independently:

- **Unsupported entry.** With 29 eligible dates, this bin's probabilities and shifts are NaN; its contributions are zero and $Q = 0, V = 0$. If the other bin is the only eligible bin left, it also has $V = 0$: the two-bin rule fails even though its shifts are finite.
- **Supported zero.** With both bins eligible, shifts $(+0.10, +0.10)$ give $Q = 0, V = 1$, distinguishing zero asymmetry from no support.
- **No matched pair.** For $\Delta = (-0.02, +0.01)$, $\Delta_\pm = \varnothing$. All weights and scores are zero and all support flags are false, even with 100 dates in each bin.

- **Heavy ties.** Request four bins from 80 values: 60 zeros and 20 ones. Linear-interpolated interior quantiles are $(0, 0, 0.25)$; filtering retains (0.25). Thus $K^{(i)} = 2$, with counts 60 and 20 if all dates are price eligible. Positions 2 and 3 are absent, bin 1 misses the count threshold, and bin 0 fails the two-bin rule. All flags are false.

Losing one horizon need not invalidate a whole score: $V = 0$ requires that no usable coordinate remains anywhere on the grid.

6.3 The synthetic-history model

The null model fits the contemporaneous mean and covariance of four observed OHLC log components, then draws independent innovations across time. It provides the reference histories for the score comparison.

Fitting and drawing log components

For the observed inputs below, suppress the path superscript (0). Component labels such as open share the superscript with the path label when both are present, as in $g_t^{\mathrm{open},(i)}$. Define

$$g_t^{\mathrm{open}} = \log\left(\frac{\mathrm{Open}_t}{\mathrm{Close}_{t-1}}\right), \qquad g_t^{\mathrm{close}} = \log\left(\frac{\mathrm{Close}_t}{\mathrm{Open}_t}\right), \tag{8}$$

$$g_t^{\mathrm{high}} = \log\left(\frac{\mathrm{High}_t}{\max(\mathrm{Open}_t, \mathrm{Close}_t)}\right), \qquad g_t^{\mathrm{low}} = \log\left(\frac{\mathrm{Low}_t}{\min(\mathrm{Open}_t, \mathrm{Close}_t)}\right). \tag{9}$$

After removing non-finite rows, let μ and Σ be the empirical mean vector and covariance matrix of these components. Add diagonal jitter before Cholesky factorization:

$$\epsilon = 10^{-12} \max(1, \max_j |\Sigma_{jj}|), \qquad L_\Sigma = \mathrm{chol}(\Sigma + \epsilon I_4).$$

Using deterministic seed `20260907`, draw row vectors

$$z_t^{(i)} \sim \mathcal{N}(0, I_4), \qquad g_t^{(i)} = z_t^{(i)} L_\Sigma^\top + \mu.$$

The draws retain the fitted contemporaneous covariance, with the stated diagonal jitter, and are independent across time.

Reconstructing a price history

Let C_{init} denote the observed close used to initialize each synthetic history, with $\mathrm{Close}_0^{(i)} = C_{\mathrm{init}}$. For synthetic steps $t \geq 1$, reconstruct

$$\mathrm{Close}_t^{(i)} = C_{\mathrm{init}} \exp\left(\sum_{j=1}^{t} (g_j^{\mathrm{open},(i)} + g_j^{\mathrm{close},(i)})\right), \tag{10}$$

$$\mathrm{Open}_t^{(i)} = \mathrm{Close}_{t-1}^{(i)} \exp(g_t^{\mathrm{open},(i)}), \tag{11}$$

$$\mathrm{High}_t^{(i)} = \max(\mathrm{Open}_t^{(i)}, \mathrm{Close}_t^{(i)}) \exp(g_t^{\mathrm{high},(i)}), \tag{12}$$

$$\mathrm{Low}_t^{(i)} = \min(\mathrm{Open}_t^{(i)}, \mathrm{Close}_t^{(i)}) \exp(g_t^{\mathrm{low},(i)}). \tag{13}$$

Observed volume is broadcast across synthetic histories. The null does not model temporal autocorrelation, volatility clustering, regime transitions, liquidity, execution, or trading costs.

Consequences for inference and OHLC consistency

The fitted mean and covariance do not reproduce temporal dependence or non-Gaussian tails. Features associated with persistent volatility or regimes can distinguish observed history from this simulator without establishing incremental predictive content. The direction and size of the score effect require measurement: symmetric changes in opposite-barrier shifts can cancel. Restricting features to those uncorrelated with a volatility proxy does not establish calibration either.

The Gaussian draw also fails to enforce $g^{\text{high}} \geq 0$ and $g^{\text{low}} \leq 0$. With Open = Close = 100, draws $g^{\text{high}} = -0.01$ and $g^{\text{low}} = +0.01$ give High $\approx$ 99.00 and Low $\approx$ 101.01, violating OHLC ordering. The reconstruction documents this limitation; it does not repair those draws. A constraint-preserving generator would be a model change requiring new validation and cache versions.

A dependent resampling or volatility model is a candidate replacement, not an automatic inferential fix. A block bootstrap can preserve both temporal dependence and the feature–future relationship under investigation. A replacement must specify which relationship its null removes, which dependence it preserves, and how calibration is checked.

6.4 Repeating measurement on each null path

For an OHLCV-derived feature, recompute its values and quantile edges independently on each synthetic path. External, calendar, and workspace-extension conditions that cannot be reconstructed from synthetic OHLCV reuse their observed feature values and observed edges on every null path.

Apply Sections 4–5 and Section 6.1 to every path. This produces $Q_k^{(1)}, \dots, Q_k^{(I)}$ at each requested bin position. The comparison matches bin positions; it does not require identical numerical quantile edges or identical effective bin counts across histories.

For a reconstructed feature, position k denotes an adaptive ordered bin with path-dependent cutoffs; ties can change its nominal quantile interval. For a fixed external feature, it denotes the same cutoffs and date group on each path. These are different conditioning policies. Applying the same policy to observed and simulated data defines a reproducible statistic, but position matching alone does not establish exchangeability or calibration. Observed volume is fixed even for reconstructed OHLCV features; its joint dynamics with synthetic prices are not preserved.

6.5 The test and decision rule

For each supported observed bin, the one-sided Monte Carlo i-value is

$$\widehat{i}_k = \frac{1 + \sum_{i=1}^{I} \mathbf{1}[Q_k^{(i)} \geq Q_k^{(0)}]}{I+1}.$$

Invalid null bins retain their zero scores in this complete ensemble. They are not dropped from the numerator or denominator. The output separately records $\sum_{i=1}^{I} V_k^{(i)}$, the number of valid null scores.

With the current $I = 1{,}000$, the denominator is 1,001. A supported observed bin clears precisely when $\widehat{i}_k < 0.05$. Unsupported observed bins are skipped. There is no family-wise or false-discovery-rate correction across bins in the current decision rule. The comparison concerns this score under the specified null model.

Support conditioning and an alternative comparison

Let $M_k = \sum_{i=1}^{I} V_k^{(i)}$ be the supported-null count. For a positive supported observed score, define

$$A_k = \sum_{i:V_k^{(i)}=1} \mathbf{1}[Q_k^{(i)} \geq Q_k^{(0)}].$$

The implemented value is $(1 + A_k)/(1 + I)$. A comparison conditional on null support would instead use

$$\widehat{\imath}_k^{\text{supported}} = \frac{1 + A_k}{1 + M_k}, \qquad M_k > 0.$$

This alternative is not the current decision rule. With no supported null paths it should be reported as untestable. Supported null zeros remain in its sample. For a supported observed zero, both comparisons equal one whenever the conditional comparison is defined.

For example, $I = 1{,}000$, $M_k = 100$, and $A_k = 4$ give implemented value $5/1{,}001 \approx 0.0050$ and conditional value $5/101 \approx 0.0495$. With $A_k = 5$, they become $6/1{,}001 \approx 0.0060$ and $6/101 \approx 0.0594$, reversing the decision. Invalid zeros lower the implemented tail rank for a fixed positive score relative to the supported comparison. This does not prove unconditional type-I error inflation: under an exchangeable null, an identically applied statistic may legitimately include a mass at zero. Conditioning on observed support changes the target population and requires its own calibration argument.

The fitted simulator does not establish that exchangeability. Its parameters are estimated from observed history, so the plus-one formula alone supplies no exact finite-sample guarantee. Resolution $1/(I + 1)$ is not a guarantee of statistical accuracy. Recommended diagnostics are M_k/I, observed and null usable-coordinate counts, effective bin counts, and both comparison values, followed by repeated full-pipeline null experiments across ties, missingness, sample sizes, and horizon grids. Currently the supported-null count, rather than this full diagnostic set, is recorded alongside scores.

Multiplicity and the meaning of clearing

For a fixed family containing m_0 true null hypotheses, valid marginal tests at level 0.05 imply an expected number of false rejections at most $0.05m_0$; independence is unnecessary for this expectation. For example, 100 features with ten true-null bin tests each would yield 50 expected false rejections if each test had size exactly 0.05. This is not a calibration result for the present simulator and does not automatically apply after conditioning on a data-selected set of supported bins. Clearing is an unadjusted exploratory flag, not a family-level error guarantee.

6.6 Worked example: applying the decision rule

To finish the running example, the observed score is $Q_k^{(0)} = 2.10$. Generate and score 1,000 null histories using the same rules. Suppose, for illustration, that 24 null scores are at least 2.10, including any ties. Then

$$\widehat{i}_k = \frac{1+24}{1{,}001} = \frac{25}{1{,}001} \approx 0.025 < 0.05.$$

This bin clears, and Stage 4 retains its complete $-0.10/0/+0.10$ shift Surface. The count of 24 is an illustrative assumption, not a simulation result: the observed shifts alone cannot determine the null comparison.

For a non-rejection example with the same positive observed score, 50 null scores at least as large give $\widehat{i}_k = 51/1{,}001 \approx 0.05095$: the bin does not clear. With 49 exceedances, $50/1{,}001 \approx 0.04995$ clears. Ties count as exceedances. An unsupported observed bin is skipped rather than assigned a passing i-value, regardless of its stored zero score.

7 Stage 4: selection and presentation

Stage 4 consumes the Stage 3 decision and introduces no new score.

7.1 Selection and rendering rules

Writing C_k for the `cleared` indicator, select exactly

$$\mathcal{K}_{\text{sel}} = \{k : C_k = \text{true}\} = \{k : V_k^{(0)} = 1 \text{ and } \widehat{i}_k < 0.05\}.$$

For every selected bin, render the complete observed shift Surface $Z_k^{(0)}$ as a signed-barrier–by–horizon heatmap.

There is no top-k truncation, family quota, effect-size gate, or representative-cell rule. Ordering by ascending $\widehat{i}_k$ and descending $Q_k^{(0)}$ is presentational only.

7.2 Worked example: selecting the shift Surface

In the running example, the observed bin is supported and its illustrative Monte Carlo i-value is $25/1{,}001 < 0.05$ (Section 6.6). Hence $C_k = \text{true}$ and $k \in \mathcal{K}_{\text{sel}}$.

Render the complete observed 5×7 shift Surface: both negative-barrier rows contain -0.10, the zero-barrier row contains 0, and both positive-barrier rows contain $+0.10$ across all seven horizons. The score $Q_k^{(0)} = 2.10$ summarizes this Surface for validation; the selected output retains all 35 entries. Selection of any other bin depends on that bin's own support and decision result.

8 Numerical and artifact contract

8.1 Surface notation and stored axes

Surface arguments (δ, θ) identify mathematical coordinates, not tensor axis positions. The following table is the storage-axis contract; no alternate scalar-subscript ordering is implied. A touch matrix is evaluated at a fixed horizon and has no horizon axis. Slicing a horizon from a logical four-axis tensor does not determine where that axis belonged. A path batch contains a subset of the paths $i = 0, 1, \ldots, I$; an observed role batch contains one path.

Quantity	Array axes, in order
OHLCV histories	path, observation, OHLCV component
Excursions at one horizon	path, observation
Touches at one horizon	path, observation, barrier
Conditional probabilities and hit counts	path, barrier, bin, horizon
Unconditional probabilities	path, barrier, horizon
Shift surfaces	path, barrier, bin, horizon
Bin observation counts	path, bin, horizon
Bin scores and validity flags	path, bin

OHLCV component order is open, high, low, close, volume. Feature values have path (or one shared path) and observation axes; edges have path (or one shared path) and edge axes; assignments have path and observation axes. A complete logical excursion tensor adds the horizon axis. The numerical computation can stream that axis without materializing every complete outcome Surface simultaneously.

Batched results retain K_{req} bin positions, with $k \geq K^{(i)}$ unsupported. Compact observed artifacts retain $K^{(0)}$ bins. The default NASDAQ configuration has $|\Delta| = 41$, $|\Theta| = 30$, $K_{\text{req}} = 10$, and $I = 1{,}000$ null replicates.

8.2 Artifacts and exact reuse

Stored artifacts

Stage 1 stores the observed conditional and unconditional probability surfaces, hit counts, and bin observation counts. Observed forward excursions, touches, and unconditional probabilities are retained as float64 source-level cache tensors. Feature values, quantile edges, and integer bin assignments are retained in the node artifact.

Shift arrays use `probability_shift` in probability-difference units. A separate shift version governs legacy conversion and invalidates old validation results without changing Stage 1 measurements. Percentage formatting is applied only in presentation.

Stage 2 stores only the shifts plus cryptographic references to its node and baseline Stage 1 artifacts. Probabilities, counts, axes, conditions, and market history are materialized from those references when required.

Shared computation and cache validity

Horizons are traversed incrementally: running future minima and maxima extend the state for shorter horizons. Synthetic histories are evaluated in memory-bounded path batches. At each batch and horizon, one touch matrix serves the baseline and all node-specific bin reductions.

Features sharing a history also share its synthetic ensemble. Their results are dependent through both the data and the common simulations. A fixed seed makes one ensemble reproducible; it does not provide independent replication across features or eliminate Monte Carlo uncertainty. Shared draws do not inherently invalidate marginal tests and can support a joint multiplicity procedure, but no such correction is currently applied.

The following quantities are reused exactly rather than approximated:

- observed forward excursions, touches, and unconditional probabilities;
- observed feature values, quantile edges, and bin assignments;

- the synthetic touch matrix shared across the baseline and all nodes in a batch; and
- common rolling feature primitives within a history or synthetic batch.

Observed caches are keyed by ordered market-history content and validated against the measurement version, barrier grid, and horizon grid. Stage 2 references are protected by SHA-256 hashes. A changed history, grid, numerical version, or referenced Stage 1 artifact invalidates reuse.